Orfalea College of Business | Cal Poly

ECON 5021 pactice R-based Unit 1 test 2026-09-16

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Please mark the boxes carefully: ☒ Not marked: ☐ or ■

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Only clearly marked and positionally accurate crosses will be processed!

Answers 1 - 15

	a	b	c	d	e
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14	☐	☐	☐	☐	☐
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	a	b	c	d	e

Answers 16 - 20

	a	b	c	d	e
16	☐	☐	☐	☐	☐
17	☐	☐	☐	☐	☐
18	☐	☐	☐	☐	☐
19	☐	☐	☐	☐	☐
20	☐	☐	☐	☐	☐
	a	b	c	d	e

1. ***This is the start of a multipart question. The information below pertains to question 1 to question 4.***

   NVDA is the ticker symbol for NVIDIA. Consider the monthly returns for this stock between March 01, 2013 and July 31, 2026.

   The following code creates a vector `x` containing the monthly returns:

```
library(quantmod)
library(dplyr)

x = getSymbols("NVDA",
               from = "2013-03-01",
               to = "2026-07-31",
               auto.assign = FALSE) %>%
  Ad() %>%
  monthlyReturn() %>%
  as.data.frame() %>%
  pull(monthly.returns)
```

   The returns are calculated using the adjusted closing price, which accounts for dividends and stock splits. `x` should contain 161 monthly returns.

   What was the maximum monthly return over this time period?

   a. 39.77%
   b. 64.39%
   c. 15.73%
   d. 38.54%
   e. None of the above

2. Given the information above.

   What was the interquartile range of monthly returns over this time period?

   a. 6.18%
   b. 14.97%
   c. 8.72%
   d. 4.61%
   e. None of the above

3. Given the information above.

   What is the 0.85-quantile of the monthly returns?

   a. 22.12%
   b. 5.06%
   c. 26.07%
   d. 21.28%
   e. None of the above

4. Given the information above.

   What is the empirical probability that the monthly return is greater than zero?

   a. 0.5584
   b. 0.3923
   c. 0.3821
   d. 0.6522
   e. None of the above

5. ***This is the start of a multipart question. The information below pertains to question 5 to question 6.***

   IBM is the ticker symbol for International Business Machines. Consider the weekly returns for this stock between August 14, 2016 and October 18, 2025.

   The following code creates a vector `x` containing the weekly returns:

```
library(quantmod)
library(dplyr)

x = getSymbols("IBM",
               from = "2016-08-14",
               to = "2025-10-18",
               auto.assign = FALSE) %>%
  Ad() %>%
  weeklyReturn() %>%
  as.data.frame() %>%
  pull(weekly.returns)
```

   The returns are calculated using the adjusted closing price, which accounts for dividends and stock splits. `x` should contain 479 weekly returns.

   We would like to compare the empirical distribution of weekly returns to a normal approximation with mean 0.003 and standard deviation 0.034.

   Focus on the 0.8-quantile. This is the return value such that approximately 80% of weekly returns are less than or equal to it. Which pair correctly reports the empirical 0.8-quantile and the 0.8-quantile from the normal approximation?

   a. The empirical 0.80-quantile is 2.51%, and the normal approximation 0.80-quantile is 3.16%
   b. The empirical 0.80-quantile is 1.44%, and the normal approximation 0.80-quantile is 3.33%
   c. The empirical 0.80-quantile is 1.63%, and the normal approximation 0.80-quantile is 3.33%
   d. The empirical 0.80-quantile is 1.16%, and the normal approximation 0.80-quantile is 4.22%
   e. None of the above

6. Given the information above.

   The CDF is defined as $F(a) = \Pr(x \leq a)$. In this context, $F(a)$ is the probability that the weekly return is less than or equal to a. Evaluate the CDF at $a = -0.01$. Which pair correctly reports the empirical probability and the probability implied by the normal approximation?

   a. The empirical probability is 0.0870, and the normal approximation probability is 0.1338
   b. The empirical probability is 0.3069, and the normal approximation probability is 0.3511
   c. The empirical probability is 0.3069, and the normal approximation probability is 0.1338
   d. The empirical probability is 0.2272, and the normal approximation probability is 0.2658
   e. None of the above

7. ***This is the start of a multipart question. The information below pertains to question 7 to question 10.***

   A balanced portfolio contains a mix of domestic stocks, international stocks, and bonds. Suppose you have a three-fund portfolio with target allocation of 25% VXF (Vanguard Extended Market ETF), 35% EWJ (iShares MSCI Japan ETF), and 40% SHV (iShares Short Treasury Bond ETF).

   The following code retrieves the weekly returns for the last three years for these three ETFs:

```
library(quantmod)
library(dplyr)

getSymbols(c("VXF","EWJ","SHV"),
           from = "2023-09-17",
           to = "2026-09-12")

weekly_return = merge(VXF %>% Ad() %>% weeklyReturn(),
                      EWJ %>% Ad() %>% weeklyReturn(),
                      SHV %>% Ad() %>% weeklyReturn())
```

Define r as the vector of weekly returns:

$$r = \begin{bmatrix} r_{\text{VXF}} \\ r_{\text{EWJ}} \\ r_{\text{SHV}} \end{bmatrix}.$$

From the weekly returns r, we construct a linear approximation to annual returns R. The question uses this annualized approximation. The mean and variance of annual returns are

$$\text{E}(R) = 52\text{E}(r), \quad \text{and} \quad \text{Var}(R) = 52\text{Var}(r).$$

Using this approximation of annual returns, what is the expected annual return of SHV in the data?

a. 4.4%
b. -7.0%
c. -0.1%
d. 12.0%
e. None of the above

8. Given the information above.

   In the data, what is the annualized standard deviation of returns for EWJ?

   a. 8.12%
   b. 15.43%
   c. 19.62%
   d. 24.66%
   e. None of the above

9. Given the information above.

   Using the target allocation weights, what is the expected annual return of the three-fund portfolio over the last three years?

   a. 13.4%
   b. 14.3%
   c. 18.0%
   d. 19.4%
   e. None of the above

10. Given the information above.

Using the target allocation weights, what is the standard deviation of annual returns for the three-fund portfolio over the last three years?

a. 6.42%
b. 9.90%
c. 5.08%
d. 10.52%
e. None of the above

11. ***This is the start of a multipart question. The information below pertains to question 11 to question 16.***

The following code retrieves monthly unemployment rate data for Alpine County, California, for the period from February 2022 through July 2026. During this period, no observation is available for 2025-10-01 because of a government shutdown. Remove this observation before conducting any analysis.

```
library(fredr)
fredr("CAALPI3URN",
       observation_start =  "2022-02-01",
       observation_end = "2026-07-01")
```

Consider the mean unemployment rate in Alpine County. Let μ_{Alpine} denote the population mean unemployment rate. Estimate this parameter using the sample average, and denote the estimator by $\hat{\mu}$.

We will assume that the variance of this estimator is

$$\text{Var}(\hat{\mu}) = \frac{\sigma^2_{Alpine}}{n},$$

where σ^2_{Alpine} is the variance of the unemployment rate for that county and n is the number of observations used in the analysis.

Using the sample variance to estimate σ^2_{Alpine}, assume that the standardized statistic follows a t distribution with 52 degrees of freedom.

What is your point estimate, $\hat{\mu}$?

a. 9.82%
b. 10.52%
c. 15.53%
d. 8.00%
e. None of the above

12. Given the information above.

What is your estimated standard error for $\hat{\mu}$?

a. 0.0304
b. 0.1700
c. 0.2244
d. 0.0136
e. None of the above

13. Given the information above.

    Suppose we wanted to conduct a two-sided hypothesis test under the null that the mean unemployment rate in Alpine County was equal to a given value, say v, i.e., $H_0 : \mu_{Alpine} = v$. Using the t distribution, what critical value should we use for a 1% significance test?

    a. 1.7127
    b. 2.6737
    c. 1.8012
    d. 3.6991
    e. None of the above

14. Given the information above.

    During this same time period, the overall unemployment rate in California was 4.95%. Given your estimate, $\hat{\mu}$, and its standard error, consider the null hypothesis that the mean unemployment rate in Alpine County was the same as the state, i.e., $H_0 : \mu_{Alpine} = 4.95$. What is the test statistic associated with this test?

    a. -0.297
    b. -21.535
    c. 0.297
    d. -13.586
    e. None of the above

15. Given the information above.

    Consider the null hypothesis, $H_0 : \mu_{Alpine} = 4.95$. Given the data, what is the strongest conclusion we can draw about this null hypothesis using a two-sided test?

    a. Reject the null at the 0.1% level
    b. Reject the null at the 1% level
    c. Reject the null at the 5% level
    d. Reject the null at the 10% level
    e. Fail to reject the null at the 10% level

16. Given the information above.

    Which of the following is the correct 99% confidence interval for your estimate?

    a. [6.9935, 9.0028]
    b. [7.0974, 8.8988]
    c. [7.1932, 8.8030]
    d. [7.3982, 8.5980]
    e. None of the above

17. ***This is the start of a multipart question. The information below pertains to question 17 to question 20.***

    In this question, you will use Monte Carlo simulation to study the sampling properties of OLS. None of the answer choices includes "None of the above," so use a sufficiently large number of Monte Carlo replications to obtain the requested level of precision.

    In this example, we relate rainfall in inches (rain) and fertilizer applied per acre (fertilizer) to crop yield in bushels per acre (yield) using the prediction equation

    $$\widehat{\text{yield}} = b_1 + b_2\text{rain} + b_3\text{fertilizer}.$$

    The following code generates one simulated data set:

```
n = 1200
rain = rgamma(n, shape = 8, scale = 5)
fertilizer = runif(n, 0, 200)
err = rnorm(n, mean = 0, sd = sqrt(656))
yield = 40 + 0.5*rain + 0.15*fertilizer + err
```

Generate many simulated data sets using this data-generating process. For each simulated data set, estimate the OLS regression and save the requested statistic. Use the resulting Monte Carlo distribution to answer the question below.

The mean of the simulated estimates of b_1 should be close to its population value of 40. Estimate the standard error of b_1 using the standard deviation of its simulated estimates. What is the estimated standard error?

a. 3.138
b. 2.561
c. 2.864
d. 3.761
e. 3.485

18. Given the information above.

    Using the simulated regressions, what is the expected value of R^2 under this data-generating process?

    a. 0.292
    b. 0.221
    c. 0.160
    d. 0.194
    e. 0.258

19. Given the information above.

    Using the simulated R^2 values, which of the following is the lower bound of the 95% interval for the sampling distribution of R^2 under this data-generating process?

    a. 0.124
    b. 0.042
    c. 0.226
    d. 0.253
    e. 0.174

20. Given the information above.

    Using the simulated R^2 values, which of the following is the upper bound of the 95% interval for the sampling distribution of R^2 under this data-generating process?

    a. 0.267
    b. 0.201
    c. 0.305
    d. 0.339
    e. 0.229